# Rebeca Hannah de Melo Oliveira, PhD


Computational Scientist in Quantitative Pharmacology

Johns Hopkins University | rhdmo94@gmail.com | LinkedIn | GitHub


## PROFESSIONAL PROFILE

Biomedical engineer and computational scientist developing mechanistic and data-informed models that connect biological mechanisms, patient heterogeneity, and therapeutic response. Expertise spans quantitative systems pharmacology, mechanistic systems biology, population pharmacokinetics, parameter estimation, uncertainty quantification, and translational simulation. Research interests include oncology, angiogenesis, vascular biology, PK/PD, and natural products.

## PROFESSIONAL EXPERIENCE

**Postdoctoral Fellow | Popel Systems Biology Lab, Johns Hopkins University**

Baltimore, MD | October 2025 - Present

- Develop mechanistic quantitative systems pharmacology models supporting oncology drug development and translational research.
- Integrate drug mechanisms, tumor biology, immune response, and biological heterogeneity within multiscale computational frameworks.
- Generate virtual populations and conduct in silico clinical-trial simulations, biomarker analyses, and model-based evaluation of treatment response.
- Collaborate across interdisciplinary academic and industry teams while communicating model assumptions, results, and translational implications.

## EDUCATION

| | | |
|---|---|---|
| **PhD, Biomedical Engineering** | Johns Hopkins University | 2025 |
| **MS, Biomedical Engineering** | University of Brasília | 2020 |
| **BS, Electrical Engineering** | University Center of Brasília | 2018 |

## CORE EXPERTISE

| | |
|---|---|
| **Modeling** | QSP; mechanistic ODE models; systems biology; PopPK and PK/PD; virtual populations |
| **Quantitative methods** | Parameter estimation; identifiability; sensitivity analysis; uncertainty quantification; simulation |
| **Computational tools** | MATLAB; SimBiology; R; Monolix; Python; scientific visualization and reproducible workflows |
| **Applications** | Oncology; angiogenesis; vascular biology; hypoxia; cerebral hemodynamics; biomedical engineering |

# Selected research experience

## MECHANISTIC AND COMPUTATIONAL MODELING

### Quantitative Systems Pharmacology in Oncology

Johns Hopkins University | 2025 - Present

- Mechanistic representation of therapeutic mechanisms and tumor-immune biology.
- Virtual population development, clinical-trial simulation, and biomarker investigation.

### Mechanistic Systems Biology of Endothelial Signaling

Johns Hopkins University | 2020 - 2025

- Developed integrative models of Notch-Dll4, HGF-VEGF, and HIF signaling in endothelial cells.
- Applied identifiability analysis, parameter estimation, global sensitivity analysis, bootstrap uncertainty quantification, and independent validation.
- Investigated angiogenesis, vascular permeability, oxygen sensing, and endothelial-cell patterning.

### Image-informed Cerebral Hemodynamics

Johns Hopkins University | 2020 - 2025

- Developed subject-specific computational approaches to predict spatiotemporal changes in cerebral blood flow.
- Integrated longitudinal in vivo imaging, vascular dynamics, mechanistic simulation, and machine learning.

### Biomedical Engineering and Translational Devices

University of Brasília and collaborating laboratories | 2016 - 2020

- Studied cardiovascular disease, peripheral arterial disease, physiological systems, biomaterials, and medical devices.
- Contributed to smart wound dressings, hypoglycemia prediction, phototherapy, and engineering models of physiological systems.

## SELECTED PEER-REVIEWED PUBLICATIONS

Oliveira, R. H. M., Pathak, A. P., & Popel, A. S. (2026). A mechanistic computational model of the HIF signaling pathway. iScience, 29(6), 116195. doi:10.1016/j.isci.2026.116195

Oliveira, R. H. M., Patil, A., Annex, B. H., Pathak, A. P., & Popel, A. S. (2025). Mechanistic computational model of HGF-VEGF-mediated endothelial cell proliferation and vascular permeability. iScience, 28(8), 113199. doi:10.1016/j.isci.2025.113199

Oliveira, R. H. M., Annex, B. H., & Popel, A. S. (2024). Endothelial cells signaling and patterning under hypoxia: a mechanistic integrative computational model including the Notch-Dll4 pathway. Frontiers in Physiology, 15, 1351753. doi:10.3389/fphys.2024.1351753

Zhang, Y., Wang, H., Oliveira, R. H. M., Zhao, C., & Popel, A. S. (2021). Systems biology of angiogenesis signaling: Computational models and omics. WIREs Mechanisms of Disease, e1550. doi:10.1002/wsbm.1550

Ochoa, M., Rahimi, R., Zhou, J., et al., including Oliveira, R. H. M. (2020). Integrated sensing and delivery of oxygen for next-generation smart wound dressings. Microsystems & Nanoengineering, 6. doi:10.1038/s41378-020-0141-7

Complete publication record: Google Scholar profile

# Academic contributions

## SELECTED HONORS AND AWARDS

**2025** Van Harreveld Memorial Award, American Physiological Society Summit

**2025** Zweifach Graduate Student Travel Award, American Physiological Society Summit

**2020** CAPES-Fulbright PhD Scholarship

**2019** Women in Science Program, British Council

**2018** Outstanding Award in the Exact and Technological Sector, University of Brasília Scientific Initiation Congress

**2015** Brazil Scientific Mobility Program - CAPES Scholarship, Purdue University

## TEACHING AND MENTORING

### Undergraduate Research Mentor

Johns Hopkins University | 2022 - 2023

- Mentored biomedical engineering students applying machine learning to peripheral arterial disease data and supported scientific presentation development.

### Head Teaching Assistant - Biological Models and Simulations

Johns Hopkins University | 2023

- Led review sessions, office hours, assessment preparation, grading coordination, and communication among teaching assistants.

### Independent Final Project Advisor

University Center of Brasília | 2019

- Advised engineering students in project development, electronic prototyping, scientific writing, and final presentations.

## SELECTED PRESENTATIONS

- An Image-Informed Computational Model for Predicting Spatiotemporal Variations in Cerebral Blood Flow - American Physiology Summit, 2025.
- An Image-Informed Computational Model for Predicting Spatiotemporal Variations in Cerebral Blood Flow - Kavli Symposium, 2025.
- Empirically-informed Computational Model of Murine Cerebral Hemodynamics - BMES Annual Meeting, 2024.
- Investigating the Effects of HGF and VEGF for Improving Angiogenesis in Peripheral Arterial Disease - Johns Hopkins DOM & WSE Research Retreat, 2024.
- Building the Endotheliome: Mechanistic Computational Systems Biology Modeling of Signaling Pathways in Endothelial Cells - NHLBI Systems Biology PI Meeting.